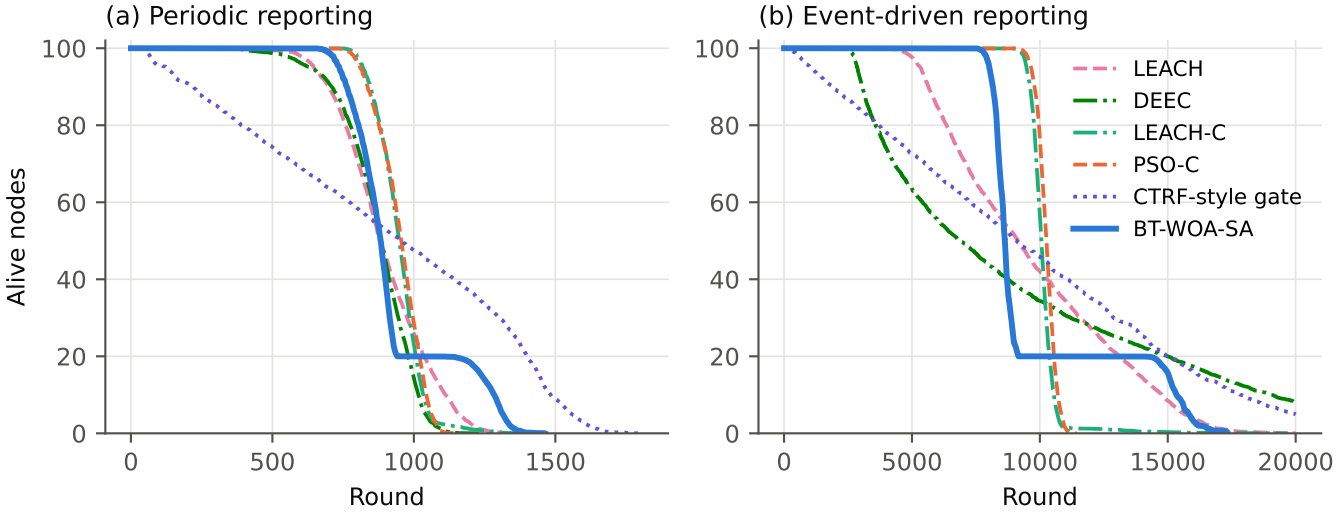

(a) Periodic reporting
(b) Event-driven reporting
Alive nodes
Round
Round
0
20
40
60
80
100
0
500
1000
1500
0
5000
10000
15000
20000
LEACH
DEEC
LEACH-C
PSO-C
CTRF-style gate
BT-WOA-SA